Smart Library Seat Management System

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Project Report: Final Stage

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1 Executive Summary

The Smart Library Seat Management System is a comprehensive web-based platform designed to optimize physical study space allocation. By introducing real-time occupancy tracking and automated policy enforcement, the system drastically reduces seat-hunting friction and ensures fair usage of library resources.

2 System Architecture

The system employs a scalable three-tier architecture:

- **Frontend:** A responsive single-page application enabling students to interact with the library floor map and manage their session statuses.
- **Backend API:** A RESTful service handling authentication, session timers, and seat allocation rules.
- **Database:** A relational database maintaining persistent state for users, physical layout maps, and historical analytics.

3 Key Features Delivered

1. **Interactive Floor Map:** Real-time visualization of seat availability across library zones.
2. **Temporary Away Permit:** Allows students to temporarily leave their seat (e.g., 30 mins limit), releasing it automatically if they overstay.
3. **Smart Seat Finder:** AI-driven recommendations based on student preferences (e.g., power outlets, quiet zones).
4. **Admin Dashboard:** Provides library administrators with occupancy metrics, allowing them to manage policies and monitor daily usage trends.

4 Evaluation & Results

The pilot testing revealed a significant reduction in the average time students spent searching for seats. Furthermore, the automated release mechanism for expired 'Temporary Away' sessions recovered a measurable percentage of seating capacity that would otherwise remain blocked by unattended belongings.

5 Conclusion & Future Work

The deployed system meets all core objectives of improving transparency and fairness in library seating. Future iterations could integrate physical IoT sensors for automated occupancy validation and construct a 3D digital twin of the library environment for enhanced spatial visualization.